Activity 5 — H. Mohammed Nawaz (BL23250G0)

Oct - 13 - 2026: Normality Detection using Q-Q plots:

Part-A

Chart 1: Resting pulse rate, n=120

* normal? NO
* Shape (from the plot): left-skewed.
* Evidence in the plot: The points curve away from reference line with a flattening at the upper end.

Chart 2: Daily study time, n=80.

* Normal? NO
* Shape (from the plot) right-skewed.
* Evidence in the plot: most points follow the reference line, but one extreme high-value point deviates sharply above it.

Chart 3: Plant height, n=40

* Normal? NO
* Shape (from the plot): Right-skewed.
* Evidence in the plot: The points shows a

strong curved pattern and deviate substantially from the reference line.

Chart 4 Resting Pulse n=40.

* Normal ? yes

* Shape (from the first) : Normal.

* evidence in the plot: The points lie approximately along the straight reference line, with only small deviations.

Part B

B1: Chart 3 depends from normality. The data appear right skewed. Try a log transformed to reduce the skewness before using mean and standard deviation methods.

B2: Fill in the blanks

```
geom_qq() + geom_qq_line().
```

complete code:

```
ggplot(df, aes(sample = value)) +
  geom_qq() +
  geom_qq_line()
```